Order Selection and Forecasting in Long Memory Functional Time Series

Interactive results for the simulation study and the empirical applications. Choose a section below to read its design, then open its results.

Overview
Simulation design
Memory estimation
Order selection
Ideal forecasting
Unified framework
Empirical study
Forecast evaluation

OVERVIEW

About this site

This site presents the simulation and empirical results for order selection and forecasting in long-memory functional time series. The framework estimates the memory parameter, fractionally differences the series to obtain a stationary representation, selects the FARMA order in this stationary domain, and produces forecasts with conformal prediction bands on the original scale.

The simulations first examine each component separately, namely memory estimation, order selection, and forecasting with known memory and order, and then evaluate the unified framework, in which estimation errors propagate through the whole procedure. The empirical study applies the framework to fertility and mortality data.

Select a section on the left to read its design, then open its results. Every results page shows the selected setting as a plot or as a table.